Results and Discussion:

The road lane detection algorithm that was implemented was put to test on several road images. The system is able to identify the white and yellow lane markings in clear images of the daytime. The HSV color space and Region of Interest (ROI), enhance the accuracy of the detection since it eliminates the unwanted backgrounds.

Original Image:

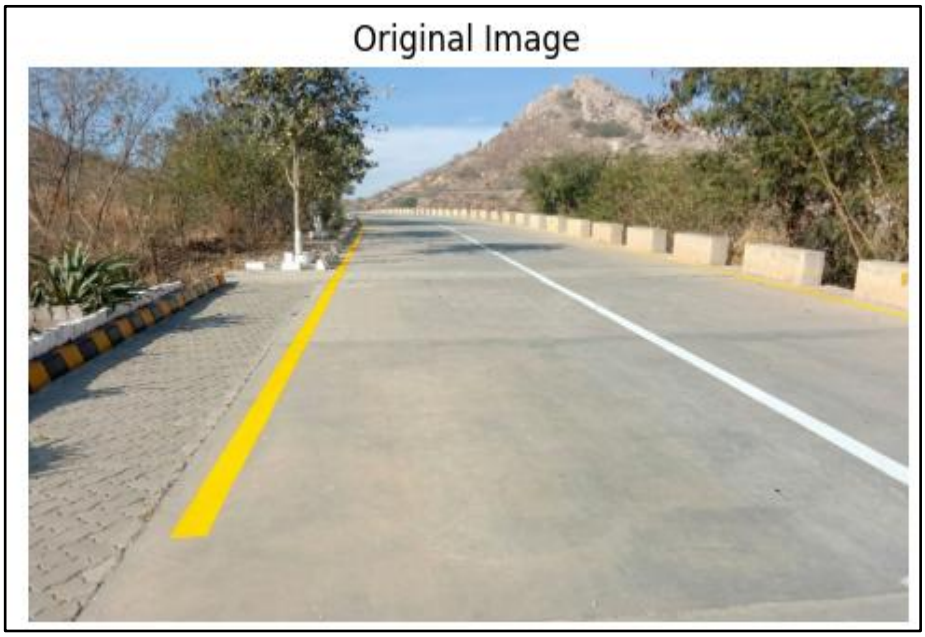


Figure 3: Original Image

Analysis of color space (Hue, Saturation, Value)

The algorithm processes the RGB image input to HSV color space that is important in identifying the lane markings.

- **Hue (H):** This differentiates between road surface and non-road in the form of trees or grass.
- **Saturation (S):** Assists in the identification of white lines as white areas lack saturation ($S < 0.3$).
- **Value (V):** Serves as a filter of brightness, showing only bright areas of the lane and disregarding the shadows.

Masking Performance:

- **White Mask:** White markings on roads are exactly recorded.
- **Yellow mask:** Identifies the yellow lanes, when they exist. A certain amount of noise might be present (e.g., grass objects, roadside objects).
- **Combined Mask:** This is a mixture of white and yellow lanes. Displayed all possible lane pixels, but can contain bright non-lane areas.

White Mask:

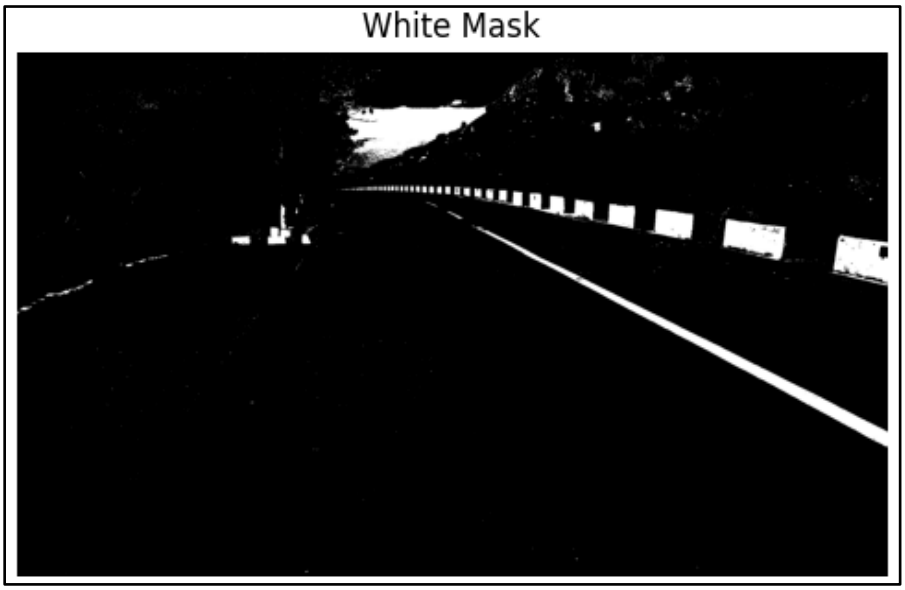


Figure 4 White Mask

Yellow Mask:

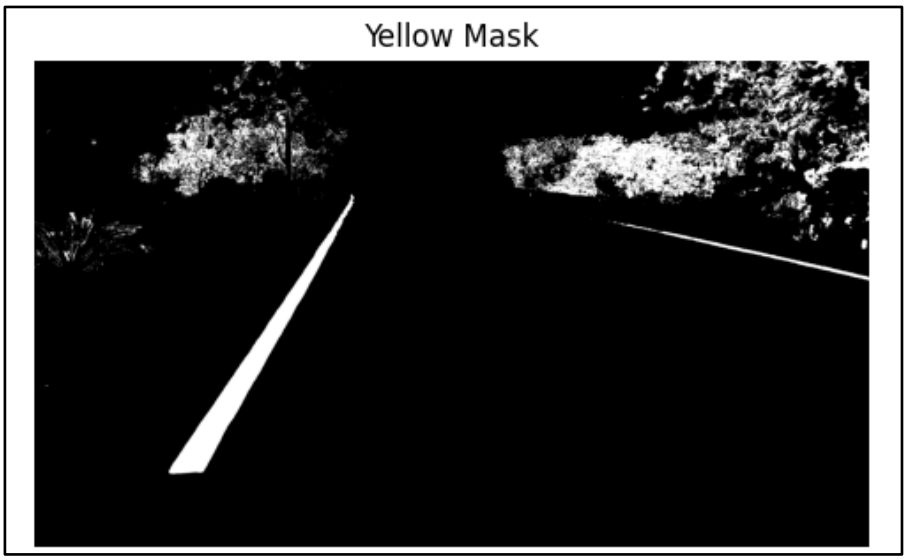


Figure 5 Yellow Mask

Combined Mask:

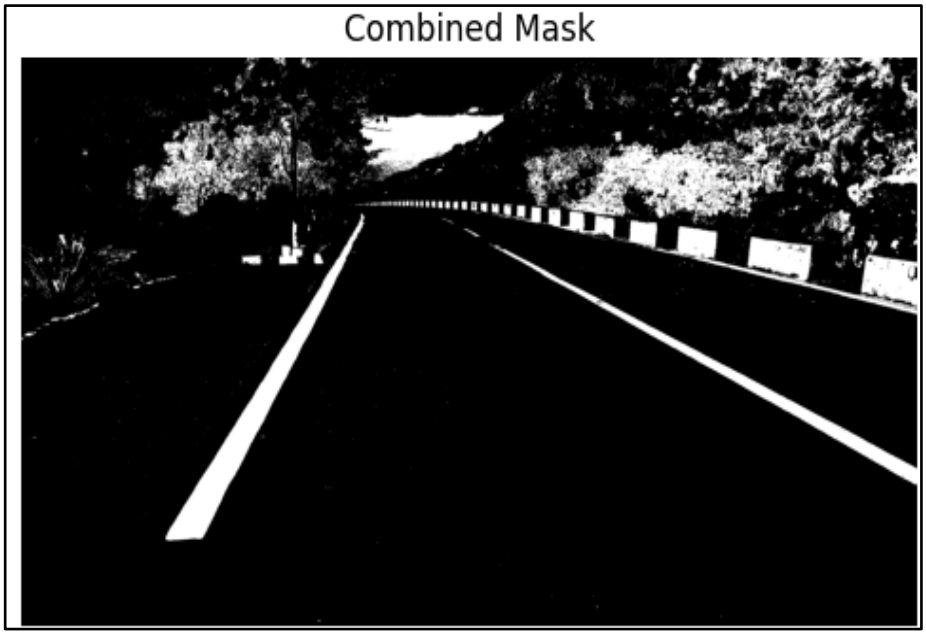


Figure 6 Combined Mask

Region of interest (ROI) Effectiveness:

The ROI Mask removes non-road features such as sky, trees and lanes.

- **Success:** Removes the majority of fake positives of the Combined Mask.
- **Output:** When the ROI masking step is implemented, the only pixels that are left are those associated with the road area.

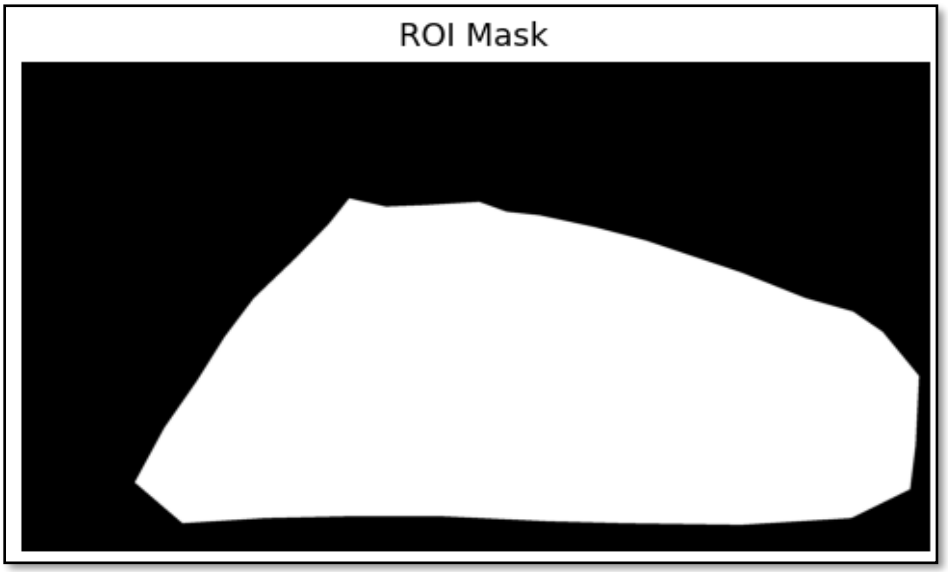


Figure 7 Region of Interest

Final Detection & Red Overlay

The end product identifies lane pixels with red color.

- **Precision:** Lane markings, such as dashed line, are easily identified.
- **Observation:** There will be some false detection on bright road barriers or shoulders in case they belong to the ROI.

Final Output:

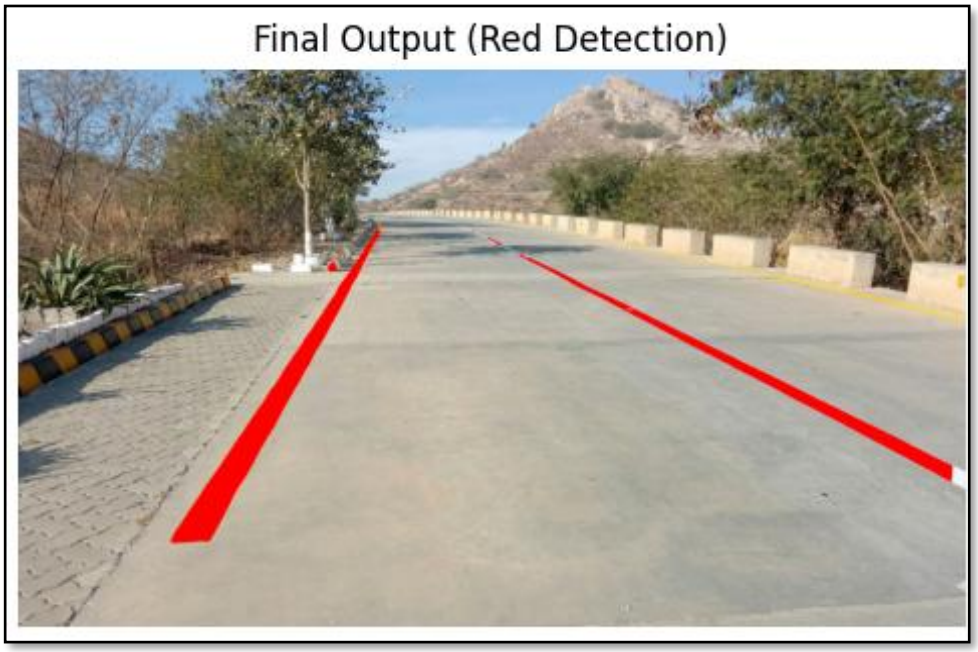


Figure 8 Red Detection Lane Output

Example of Unsuccessful Detection

Some images produce incorrect results due to challenging conditions:

- **Night-time or low-light image** – lane markings are not visible.
- **Shadowed roads** – bright areas under shadows can cause false detection.

Original Picture with shadow:



Figure 9 Original Picture with shadow

Final Output:



Figure 10 Final Output